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import javax.swing.*;
import javax.swing.border.*;
import javax.swing.table.*;
import java.awt.*;
import java.awt.event.*;
import java.net.MalformedURLException;
import java.net.URL;
import java.util.ArrayList;
import java.util.Random;

public class SEUModernPro extends JFrame {

    private final Color BG_DARK = new Color(30, 41, 59);
    private final Color BG_LIGHT = new Color(241, 245, 249);
    private final Color CARD_BG = Color.WHITE;
    private final Color ACCENT_BLUE = new Color(59, 130, 246);
    private final Color TEXT_PRIMARY = new Color(51, 65, 85);
    private final Color TEXT_SECONDARY = new Color(148, 163, 184);

    private final Font MAIN_FONT = new Font("Segoe UI", Font.PLAIN, 14);
    private final Font TITLE_FONT = new Font("Segoe UI", Font.BOLD, 24);
    private final Font NAV_FONT = new Font("Segoe UI", Font.PLAIN, 15);

    private JPanel contentPanel;
    private JLabel headerTitle;
    private final ArrayList<Student> studentList = new ArrayList<>();
    private JButton activeNavButton = null;

    public SEUModernPro() {
        setTitle("SEU | Student Success Portal v3.7");
        setSize(1280, 800);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout());

        populateRandomStudents(100);

        initUI();
    }

    private void populateRandomStudents(int count) {
        String[] firstNames = {"Abir", "Sadia", "Tanvir", "Nusrat", "Arif", "Fariha", "Kamal", "Laila",
            "Zahid", "Mitu"};

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String[] lastNames = {"Ahmed", "Hasan", "Khan", "Islam", "Rahman", "Jahan", "Akter",
"Uddin", "Chowdhury", "Talukder"};
String[] depts = {"CSE", "BBA", "EEE", "Economics", "English", "Textile", "Pharmacy"};
String[] subjs = {"Software Engineering", "Finance", "Power Systems", "Microeconomics",
"Literature", "Merchandising", "Clinical Pharmacy"};
String[] quals = {"HSC", "A-Level", "Diploma"};
String[] cities = {"Dhaka", "Chittagong", "Sylhet", "Rajshahi", "Khulna", "Barisal"};

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Random rand = new Random();
for (int i = 0; i < count; i++) {
    String name = firstNames[rand.nextInt(firstNames.length)] + " " +
lastNames[rand.nextInt(lastNames.length)];
    String address = cities[rand.nextInt(cities.length)];
    int deptIdx = rand.nextInt(depts.length);
    String dept = depts[deptIdx];
    String subj = subjs[deptIdx];
    int year = 2020 + rand.nextInt(5);
    String qual = quals[rand.nextInt(quals.length)];

    studentList.add(new Student(name, address, dept, subj, year, qual));
}
}

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private void initUI() {
    JPanel sidebar = new JPanel();
    sidebar.setLayout(new BoxLayout(sidebar, BoxLayout.Y_AXIS));
    sidebar.setPreferredSize(new Dimension(260, 0));
    sidebar.setBackground(BG_DARK);
    sidebar.setBorder(new EmptyBorder(25, 20, 25, 20));

    JLabel brand = new JLabel("<html><span style='color:#3b82f6;
font-size:1.2em'>SEU</span> PORTAL</html>");
    brand.setFont(new Font("Segoe UI", Font.BOLD, 22));
    brand.setForeground(Color.WHITE);
    brand.setAlignmentX(Component.LEFT_ALIGNMENT);
    sidebar.add(brand);
    sidebar.add(Box.createVerticalStrut(60));

    sidebar.add(createNavSectionLabel("MAIN MENU"));
    sidebar.add(Box.createVerticalStrut(15));

    JButton btnHome = createNavButton("<img alt='home icon' data-bbox='465 868 485 888'> Dashboard", e -> showDashboard());
    JButton btnSearch = createNavButton("<img alt='search icon' data-bbox='465 888 485 908'> Search Student", e -> showSearch());
}

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JButton btnList = createNavButton("👥 Directory", e -> showDatabase());
JButton btnInfo = createNavButton("🏠 Campus Details", e -> showCampus());
JButton btnAdd = createNavButton("🌟 New Admission", e -> admissionFlow());

sidebar.add(btnHome); sidebar.add(btnSearch); sidebar.add(btnList);
sidebar.add(btnInfo); sidebar.add(btnAdd);

add(sidebar, BorderLayout.WEST);

JPanel mainArea = new JPanel(new BorderLayout());
mainArea.setBackground(BG_LIGHT);

JPanel header = new JPanel(new BorderLayout());
header.setBackground(CARD_BG);
header.setPreferredSize(new Dimension(0, 70));
header.setBorder(new CompoundBorder(new MatteBorder(0, 0, 1, 0, new Color(226, 232, 240)), new EmptyBorder(0, 40, 0, 40)));

headerTitle = new JLabel("Dashboard");
headerTitle.setFont(TITLE_FONT);
headerTitle.setForeground(TEXT_PRIMARY);
header.add(headerTitle, BorderLayout.WEST);
mainArea.add(header, BorderLayout.NORTH);

contentPanel = new JPanel(new BorderLayout());
contentPanel.setBackground(BG_LIGHT);
contentPanel.setBorder(new EmptyBorder(40, 40, 40, 40));
mainArea.add(contentPanel, BorderLayout.CENTER);

add(mainArea, BorderLayout.CENTER);
setActiveButton(btnHome);
showDashboard();
}

private void showSearch() {
    headerTitle.setText("Search Student");
    JPanel wrapper = new JPanel(new BorderLayout());
    wrapper.setOpaque(false);

    JPanel searchBarPanel = new JPanel(new FlowLayout(FlowLayout.LEFT, 0, 0));
    searchBarPanel.setOpaque(false);
    JTextField searchField = new JTextField(20);
    searchField.setPreferredSize(new Dimension(300, 45));

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searchField.setFont(MAIN_FONT);
searchField.setBorder(BorderFactory.createCompoundBorder(new LineBorder(new
Color(203, 213, 225), 1), new EmptyBorder(5, 15, 5, 15)));
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```
JButton btnGo = new JButton("Search ID");
btnGo.setPreferredSize(new Dimension(120, 45));
btnGo.setBackground(Color.BLACK);
btnGo.setForeground(Color.WHITE);
btnGo.setFont(new Font("Segoe UI", Font.BOLD, 14));
btnGo.setFocusPainted(false);
btnGo.setBorderPainted(false);
btnGo.setCursor(new Cursor(Cursor.HAND_CURSOR));
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searchBarPanel.add(searchField);
searchBarPanel.add(Box.createHorizontalStrut(10));
searchBarPanel.add(btnGo);
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JPanel resultArea = new JPanel();
resultArea.setLayout(new BoxLayout(resultArea, BoxLayout.Y_AXIS));
resultArea.setOpaque(false);
resultArea.setBorder(new EmptyBorder(30, 0, 0, 0));
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ActionListener performSearch = e -> {
    resultArea.removeAll();
    String input = searchField.getText().trim();
    if (input.isEmpty()) { JOptionPane.showMessageDialog(this, "Please enter an ID.");
return; }
    try {
        int id = Integer.parseInt(input);
        Student found = null;
        for (Student s : studentList) {
            if (s.getId() == id) { found = s; break; }
        }
        if (found != null) {
            resultArea.add(createStudentProfileCard(found));
        } else {
            JLabel notFound = new JLabel("No student found with ID: " + id);
            notFound.setFont(MAIN_FONT); notFound.setForeground(Color.RED);
            resultArea.add(notFound);
        }
    } catch (NumberFormatException ex) { JOptionPane.showMessageDialog(this, "ID must
be a number."); }
    resultArea.revalidate(); resultArea.repaint();
}
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};

btnGo.addActionListener(performSearch);
wrapper.add(searchBarPanel, BorderLayout.NORTH);
wrapper.add(resultArea, BorderLayout.CENTER);
updateView(wrapper);
}

void admissionFlow() {
    UIManager.put("OptionPane.background", CARD_BG);
    UIManager.put("Panel.background", CARD_BG);

    String[] schools = {"School of Engineering (SE)", "School of Arts & Social Sciences (SASS)", "School of Business Studies (SBS)"};
    String selectedSchool = (String) JOptionPane.showInputDialog(this, "Select Faculty:", "Step 1: Faculty Selection", JOptionPane.PLAIN_MESSAGE, null, schools, schools[0]);
    if (selectedSchool == null) return;

    String[] depts;
    if (selectedSchool.contains("(SE)")) depts = new String[]{"CSE", "EEE", "Textile Engineering", "Architecture"};
    else if (selectedSchool.contains("(SASS)")) depts = new String[]{"Economics", "English", "Bangla"};
    else depts = new String[]{"Business Administration"};

    String selectedDept = (String) JOptionPane.showInputDialog(this, "Select Department:", "Step 2: Department Selection", JOptionPane.PLAIN_MESSAGE, null, depts, depts[0]);
    if (selectedDept == null) return;

    JTextField nameF = new JTextField();
    JTextField addrF = new JTextField();
    JTextField yearF = new JTextField();
    JTextField qualF = new JTextField();
    JTextField subjF = new JTextField();

    Object[] form = {"Full Name:", nameF, "Address:", addrF, "Year:", yearF, "Qualification:", qualF, "Subject:", subjF};
    int result = JOptionPane.showConfirmDialog(this, form, "Step 3: Personal Information", JOptionPane.OK_CANCEL_OPTION, JOptionPane.PLAIN_MESSAGE);
    if (result == JOptionPane.OK_OPTION) {
        try {
            Student s = new Student(nameF.getText(), addrF.getText(), selectedDept, subjF.getText(), Integer.parseInt(yearF.getText()), qualF.getText());
            studentList.add(s);
            JOptionPane.showMessageDialog(this, "Student Admitted Successfully!\nAssigned

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ID: " + s.getId());
    } catch (HeadlessException | NumberFormatException e) {
OptionPane.showMessageDialog(this, "Validation Failed: Ensure Year is numeric."); }
    }
}

private JPanel createStudentProfileCard(Student s) {
    JPanel card = new JPanel(new GridLayout(0, 1, 8, 8));
    card.setBackground(CARD_BG);
    card.setBorder(new CompoundBorder(new LineBorder(new Color(226, 232, 240), 1), new
EmptyBorder(25, 25, 25, 25)));
    card.setMaximumSize(new Dimension(650, 300));
    card.setAlignmentX(Component.LEFT_ALIGNMENT);
    String bold = "<html><b style='color:#64748b;'>";
    card.add(new JLabel(bold + "Name: </b>" + s.getName()));
    card.add(new JLabel(bold + "ID: </b>" + s.getId()));
    card.add(new JLabel(bold + "Department: </b>" + s.getDept()));
    card.add(new JLabel(bold + "Subject: </b>" + s.getSubj()));
    card.add(new JLabel(bold + "Admission Year: </b>" + s.getYear()));
    card.add(new JLabel(bold + "Qualification: </b>" + s.getQual()));
    card.add(new JLabel(bold + "Address: </b>" + s.getAddr()));
    return card;
}

private void showDashboard() {
    headerTitle.setText("Portal Overview");
    JPanel dashGrid = new JPanel(new GridLayout(1, 3, 30, 0)); dashGrid.setOpaque(false);
    dashGrid.add(createSummaryCard("Students", String.valueOf(studentList.size()),
ACCENT_BLUE, "🎓"));
    dashGrid.add(createSummaryCard("Faculties", "3", new Color(16, 185, 129), "👥"));
    dashGrid.add(createSummaryCard("Status", "Online", new Color(245, 158, 11), "🌐"));
    updateView(dashGrid);
}

private void showDatabase() {
    headerTitle.setText("Student Directory");
    String[] cols = {"ID", "Name", "Department", "Year"};
    Object[][] data = new Object[studentList.size()][4];
    for(int i=0; i<studentList.size(); i++) {
        Student s = studentList.get(i);
        data[i] = new Object[]{s.getId(), s.getName(), s.getDept(), s.getYear()};
    }
    JTable table = new JTable(new DefaultTableModel(data, cols));

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        table.setRowHeight(35);
        updateView(new JScrollPane(table));
    }

    private void showCampus() {
        headerTitle.setText("Campus Details");
        try {
            URL url = new URL("https://cubeinsidebd.com/media/images/SEU-11.width-1200.jpg");
            JLabel img = new JLabel(new ImageIcon(new
            ImageIcon(url).getImage().getScaledInstance(800, 450, Image.SCALE_SMOOTH)));
            updateView(img);
        } catch (MalformedURLException e) { updateView(new JLabel("Check Connection to Load
            Image")); }
    }

    private void updateView(JComponent component) {
        contentPanel.removeAll();
        contentPanel.add(component, BorderLayout.CENTER);
        contentPanel.revalidate(); contentPanel.repaint();
    }

    private void setActiveButton(JButton active) {
        if (activeNavButton != null) { activeNavButton.setBackground(BG_DARK);
            activeNavButton.setForeground(TEXT_SECONDARY); }
        activeNavButton = active; activeNavButton.setBackground(new Color(51, 65, 85));
        activeNavButton.setForeground(Color.WHITE);
    }

    private JButton createNavButton(String text, ActionListener action) {
        JButton btn = new JButton(text);
        btn.setFont(NAV_FONT); btn.setForeground(TEXT_SECONDARY);
        btn.setBackground(BG_DARK);
        btn.setBorderPainted(false); btn.setFocusPainted(false);
        btn.setHorizontalAlignment(SwingConstants.LEFT);
        btn.setMaximumSize(new Dimension(220, 50)); btn.setCursor(new
        Cursor(Cursor.HAND_CURSOR));
        btn.setBorder(new EmptyBorder(10, 15, 10, 0));
        btn.addActionListener(e -> { setActiveButton(btn); action.actionPerformed(e); });
        return btn;
    }

    private JLabel createNavSectionLabel(String text) {
        JLabel lbl = new JLabel(text);
    }

```

```

        lbl.setFont(new Font("Segoe UI", Font.BOLD, 11)); lbl.setForeground(new Color(100, 116,
139)); return lbl;
    }

```

```

    private JPanel createSummaryCard(String title, String value, Color accentColor, String
iconEmoji) {
        JPanel card = new JPanel(new BorderLayout());
        card.setBackground(CARD_BG); card.setBorder(new LineBorder(new Color(226, 232,
240), 1));
        JPanel content = new JPanel(); content.setLayout(new BoxLayout(content,
BoxLayout.Y_AXIS)); content.setOpaque(false);
        content.setBorder(new EmptyBorder(20, 25, 20, 25));
        content.add(new JLabel(iconEmoji) {{ setFont(new Font("Segoe UI Emoji", Font.PLAIN,
24)); }});
        content.add(new JLabel(title) {{ setForeground(TEXT_SECONDARY); }});
        content.add(new JLabel(value) {{ setFont(new Font("Segoe UI", Font.BOLD, 28)); }});
        JPanel accentStrip = new JPanel(); accentStrip.setBackground(accentColor);
accentStrip.setPreferredSize(new Dimension(0, 4));
        card.add(accentStrip, BorderLayout.NORTH);
        card.add(content, BorderLayout.CENTER); return card;
    }

```

```

    public static void main(String[] args) {
        try { UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
        } catch (ClassNotFoundException | IllegalAccessException | InstantiationException |
UnsupportedLookAndFeelException e) {}
        SwingUtilities.invokeLater(() -> new SEUModernPro().setVisible(true));
    }
}

```

```

abstract class Person {
    protected String name, address;
    Person(String name, String address) {
        this.name = name;
        this.address = address;
    }
    public abstract String getName();
    public abstract String getAddr();
}

```

```

class Student extends Person {
    private static int counter = 20261000;
    private final int id;

```



```
private int year;  
private String department, subject, qualification;
```

```
Student(String n, String a, String d, String s, int y, String q) {  
    super(n, a);  
    this.id = ++counter;  
    this.department = d;  
    this.subject = s; this.year = y; this.qualification = q;  
}
```

```
public int getId() { return id; }
```

```
@Override  
public String getName() { return name; }
```

```
@Override  
public String getAddr() { return address; }
```

```
public String getDept() { return department; }  
public String getSubj() { return subject; }  
public int getYear() { return year; }  
public String getQual() { return qualification; }  
}
```